FHIR INTEGRATION PROSPECTS

Utilizing HL7 FHIR R4 Standards in the Smartband V001 Project

A Roadmap for Health Data Interoperability

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Submitted to: Prof. Dr. Javidi da Costa, Joao Paulo
Date: 2nd June 2026

1. Introduction

The Smartband V001 project, developed at Hochschule Hamm-Lippstadt (HSHL), aims to produce a wearable health monitoring device capable of collecting, transmitting, and displaying physiological data. As the project matures toward clinical and research applicability, the integration of internationally recognized health data standards becomes not only desirable but essential [1].

This document outlines the prospects and planned roadmap for integrating HL7 FHIR R4 (Fast Healthcare Interoperability Resources) standards into the Smartband V001 ecosystem. It describes the motivation, applicable FHIR resource types, sensor requirements, planned system architecture, and the phased implementation plan that will guide future development [2].

2. What is HL7 FHIR?

HL7 FHIR (Fast Healthcare Interoperability Resources) is an international standard published by Health Level Seven International (HL7) for exchanging electronic health records [3]. FHIR defines a set of modular components called Resources — structured data objects that represent clinical concepts such as patients, observations, medications, and diagnoses.

FHIR R4 (Release 4) is the current stable version and is widely adopted by hospitals, health apps, insurance systems, and government health platforms globally [4]. Its REST-based API design makes it particularly well-suited for integration with IoT devices such as wearables [5].

2.1 Key FHIR Concepts

Concept	Description	Relevance to Smartband V001
Resource	A modular data object representing a clinical concept	Each sensor reading becomes an Observation resource
Bundle	A container grouping multiple resources together	All vitals sent as one Bundle per reading cycle
Patient	Represents the person wearing the device	Each Smartband user has a FHIR Patient resource

Concept	Description	Relevance to Smartband V001
Observation	Represents a single measured value (HR, Temp, etc.)	Core resource for all Smartband sensor data
LOINC	Logical Observation Identifiers Names and Codes — coding system	Used to code each sensor measurement internationally
UCUM	Unified Code for Units of Measure	Standardizes units (bpm, Cel, [g], kcal)
REST API	HTTP-based interface for accessing FHIR resources	Smartband server exposes /api/fhir/ endpoints

3. Why FHIR for Smartband V001?

Wearable health devices generate continuous streams of physiological data. Without a standard format, this data remains siloed — it cannot be consumed by hospital systems, shared between healthcare providers, or compared across different devices [6]. FHIR solves this by providing a universal language for health data.

3.1 Core Motivations

- Interoperability — FHIR-compliant data can be consumed by any EHR system, hospital platform, or health app that supports the standard, without custom integration work [3].
- Clinical Validity — Using LOINC codes and UCUM units ensures that sensor readings are interpreted correctly by clinicians and automated systems alike [7].
- Research Applicability — FHIR-formatted data can be directly imported into clinical research platforms and data analysis tools [8].
- Regulatory Alignment — Many healthcare regulations and certification frameworks (MDR in Europe, FDA in the US) increasingly require or encourage FHIR compliance for medical devices [9].
- Scalability — A FHIR-based architecture scales naturally from a single patient to a hospital-wide deployment of thousands of devices [4].
- Doctor-Patient Workflow — FHIR enables structured clinical workflows where doctors can view, annotate, and act on patient data through standardized interfaces [5].

4. Sensor Requirements and FHIR Resource Mapping

The following table maps each planned sensor on the Smartband V001 to its corresponding FHIR Observation resource, LOINC code, and UCUM unit [7]. This mapping defines the data contract that the device will expose via its FHIR API.

Sensor	Measurement	Hardware	LOINC Code	UCUM Unit	Status
Optical HR Sensor	Heart Rate	MAX30102	8867-4	/min	Planned — on PCB
Optical HR Sensor	SpO2 (Blood Oxygen)	MAX30102	2708-6	%	Planned — on PCB
Temperature Sensor	Body Temperature	MLX90632	8310-5	Cel	Planned — on PCB
Accelerometer/Gyro	Acceleration	MPU-6050	80493-0	[g]	Planned — on PCB
Accelerometer/Gyro	Step Count	MPU-6050	55423-8	steps	Calculated
Accelerometer/Gyro	Activity Level	MPU-6050	77592-7	score	Calculated
HR + Steps	Calories Burned	Calculated	41981-2	kcal	Calculated
HR + Demographics	Heart Rate Variability	MAX30102	80404-7	ms	Future
Future — GSR Sensor	Stress Level	GSR/EDA	74767-6	score	Future hardware
Future — BP Sensor	Blood Pressure	Cuff/PPG	55284-4	mm[Hg]	Future hardware

4.1 Future Sensor Additions

The following sensors are not currently on the PCB but are planned for future hardware revisions. Each sensor expands the FHIR Observation coverage and clinical value of the device [13].

Sensor	Measurement	LOINC Code	Clinical Value
GSR / EDA Sensor	Galvanic Skin Response (Stress)	74767-6	Mental health and stress monitoring
Blood Pressure Module	Systolic / Diastolic BP	55284-4	Cardiovascular health monitoring
ECG Module	Electrocardiogram	11524-6	Cardiac arrhythmia detection
Blood Glucose Sensor	Blood Glucose Level	15074-8	Diabetes management
Respiratory Rate Sensor	Breathing Rate	9279-1	Respiratory health monitoring

GSR / EDA Sensor — Galvanic Skin Response

A Galvanic Skin Response (GSR) or Electrodermal Activity (EDA) sensor measures changes in skin conductance caused by sweat gland activity, which correlates with emotional arousal and stress [14]. Recommended module: Grove GSR Sensor v1.2 (Seeed Studio SKU 101020052) or AD8232-based EDA breakout. Interface: Analog ADC. Supply: 3.3V. Requires two skin contact electrodes on the wrist. Would provide FHIR Observation for Stress Level (LOINC 74767-6).

Blood Pressure Module

Wearable blood pressure sensing can be achieved via PPG (Photoplethysmography) waveform analysis from the MAX30102 combined with calibration, or via a dedicated module such as the Omron MicroBP or the BP2 module from ProtoCentral. Would provide FHIR Observations for Systolic and Diastolic Blood Pressure (LOINC 55284-4, unit: mm[Hg]) [15]. Critical for cardiovascular disease monitoring and hypertension management.

ECG Module

The AD8232 (Analog Devices) is a single-lead ECG front-end chip widely available as a breakout module [16]. Recommended module: SparkFun Single Lead Heart Rate Monitor AD8232 (DEV-12650) or similar. Interface: Analog ADC. Supply: 3.3V. Requires two to three electrode contact points (RA, LA, RL). Would provide FHIR Observation for Electrocardiogram (LOINC 11524-6). Can detect arrhythmia, atrial fibrillation, and other cardiac events.

Blood Glucose Sensor

Non-invasive continuous glucose monitoring (CGM) using near-infrared spectroscopy is an emerging wearable technology. For a prototype, the Texas Instruments AFE4490 optical front-end paired with NIR LEDs can be evaluated [17]. Alternatively, invasive CGM patches (e.g. Abbott FreeStyle Libre chip) can provide interstitial glucose. Would provide FHIR Observation for Blood Glucose Level (LOINC 15074-8, unit: mmol/L). High clinical demand for CGM in wearables for diabetes management.

Respiratory Rate Sensor

Respiratory rate can be derived from the existing MPU-6050 accelerometer data (chest movement analysis) or from the MAX30102 PPG signal (respiratory sinus arrhythmia). For a dedicated sensor, the MEMS-based TDK InvenSense ICM-42688-P provides higher precision motion data suitable for breathing pattern analysis [18]. Would provide FHIR Observation for Breathing Rate (LOINC 9279-1, unit: /min). Elevated respiratory rate is an early indicator of sepsis and critical illness.

5. Planned FHIR System Architecture

The planned FHIR integration architecture for the Smartband V001 ecosystem consists of four layers, each building on the previous to create a complete health data pipeline from sensor to clinical system [5].

Layer	Component	Responsibility
Layer 1 — Sensing	Band01_Rev1 Hardware	Collect raw physiological data from sensors via I2C/SPI
Layer 2 — Processing	CC2640R2F Firmware	Process raw data, apply calibration, transmit via BLE
Layer 3 — Gateway	Raspberry Pi 5 + Flask Server	Receive BLE data, structure as FHIR R4 resources, expose REST API
Layer 4 — Consumption	Hospital EHR / Phone App	Consume FHIR API, display to clinicians and patients

The key transformation happens at Layer 3. Raw sensor values arriving via BLE are mapped to FHIR Observation resources with proper LOINC codes and UCUM units before being stored and exposed via the REST API. This means any downstream consumer — regardless of vendor — can interpret the data correctly without custom mapping [4].

5.2 Planned FHIR API Endpoints

The Smartband V001 gateway server will expose the following FHIR R4 compliant REST endpoints [3]:

Endpoint	Resource Type	Description
/api/fhir/patient	Patient Bundle	All registered patients with demographics
/api/fhir/patient/{id}	Patient	Individual patient resource by ID
/api/fhir/observation	Bundle	All current vital observations for active patient
/api/fhir/observation/hr	Observation	Heart Rate (LOINC 8867-4)
/api/fhir/observation/spo2	Observation	Blood Oxygen (LOINC 2708-6)
/api/fhir/observation/temp	Observation	Body Temperature (LOINC 8310-5)
/api/fhir/observation/acc	Observation	Acceleration (LOINC 80493-0)
/api/fhir/observation/steps	Observation	Step Count (LOINC 55423-8)
/api/fhir/observation/calories	Observation	Calories Burned (LOINC 41981-2)
/api/fhir/observation/history	Bundle	Historical observation bundle (last N readings)
/api/fhir/condition	Condition	Detected health conditions / alerts
/api/fhir/practitioner	Practitioner	Assigned healthcare provider
/api/fhir/careplan	CarePlan	Patient monitoring and care instructions

6. Planned Clinical Workflow

A key goal of the FHIR integration is to enable a meaningful clinical workflow where healthcare providers can monitor patients remotely, receive alerts, and document clinical decisions — all through a standards-based interface [6].

6.1 Patient Monitoring Workflow

1. Patient wears Smartband V001 which continuously collects HR, temperature, SpO2, and acceleration data.

2. Firmware on CC2640R2F transmits data via BLE to the Raspberry Pi gateway.
3. Gateway structures data as FHIR Observation resources and stores them.
4. Doctor opens the web app or connects their EHR system to the FHIR API.
5. Doctor views live vitals, historical trends, and alert history for each patient.
6. Doctor updates patient status (Normal / Under Observation / Critical) and adds clinical notes.
7. Status updates are stored as FHIR Condition resources and reflected immediately on all connected devices.
8. Emergency alerts are pushed via FHIR Subscription to the doctor's device when thresholds are exceeded.

6.2 FHIR Resources Used in Clinical Workflow

FHIR Resource	Used For	When
Observation	Each sensor reading	Every 2 seconds continuously
Patient	Patient demographics and device assignment	On registration and updates
Condition	Diagnosed or detected health conditions	When doctor adds a clinical note
Practitioner	Doctor details and credentials	On doctor login
CarePlan	Monitoring plan and threshold instructions	Set by doctor per patient
AllergyIntolerance	Patient allergies relevant to care	During patient registration
AuditEvent	Log of all accesses and changes for compliance	Continuously for audit trail

7. Phased Implementation Plan

The FHIR integration will be implemented in phases, with each phase building on the previous. This approach allows the project to demonstrate incremental progress while managing complexity [2].

Phase	Focus	Key Deliverables	Timeline
Phase 1	Foundation	FHIR Patient and Observation resources, LOINC coding, REST API endpoints, basic phone app	Current phase
Phase 2	Real Sensor Data	Activate firmware sensor reads, replace simulated data with real	After PCB assembly

Phase	Focus	Key Deliverables	Timeline
		MAX30102/MPU-6050/MLX90632 readings via BLE	
Phase 3	Clinical Workflow	FHIR Condition, CarePlan, Practitioner resources, doctor dashboard with full FHIR backend, FHIR Subscription alerts	Following Phase 2
Phase 4	EHR Integration	Connect to external FHIR server (HAPI FHIR), validate with FHIR validator, integrate with hospital EHR system	Advanced phase
Phase 5	Certification	Evaluate compliance with EU MDR, GDPR data handling, HL7 FHIR conformance testing, ISO 13485	Final phase

8. FHIR Server Options

For production deployment, the Smartband V001 system will need to connect to a dedicated FHIR server [4]. The following options are being considered:

Server	Type	Cost	Best For
HAPI FHIR	Open source Java server	Free	Self-hosted production deployment, full FHIR R4 compliance
Azure Health Data	Cloud FHIR service	Pay per use	Cloud-scale deployment with Azure integration
AWS HealthLake	Cloud FHIR service	Pay per use	Amazon ecosystem, ML-ready health data lake
Google Cloud FHIR	Cloud FHIR service	Pay per use	BigQuery integration for research analytics
Firely Server	Commercial FHIR server	Licensed	Enterprise deployment with full support
SmileCDR	Commercial FHIR platform	Licensed	Hospital-grade deployment with compliance tools

For the academic project scope, HAPI FHIR running locally on a server or cloud VM is the recommended option due to its open-source nature and full FHIR R4 compliance [4].

9. Data Privacy and Compliance Considerations

Health data is among the most sensitive categories of personal data under GDPR in the European Union [9]. Any FHIR implementation for Smartband V001 must address the following compliance requirements:

Requirement	Standard / Regulation	Planned Approach
Data Encryption in Transit	GDPR Article 32, TLS	HTTPS/TLS for all FHIR API communication
Data Encryption at Rest	GDPR Article 32	Encrypted storage for all patient data on server
Patient Consent	GDPR Article 7	Explicit consent recorded as FHIR Consent resource
Right to Erasure	GDPR Article 17	FHIR DELETE operations for patient data removal
Access Control	GDPR Article 25, HIPAA	OAuth 2.0 / SMART on FHIR authentication
Audit Logging	ISO 27001, HIPAA	FHIR AuditEvent resources for all data access
Data Minimization	GDPR Article 5	Collect only clinically relevant sensor data
Medical Device Regulation	EU MDR 2017/745	Evaluate classification and conformity requirements

10. SMART on FHIR — Authentication and Authorization

SMART on FHIR (Substitutable Medical Applications, Reusable Technologies) is the standard authorization framework built on top of FHIR [3]. It defines how applications authenticate users and request access to health data in a secure, standardized way.

For the Smartband V001 project, SMART on FHIR is planned for Phase 3 and will enable:

- Doctor login with role-based access — doctors can only see their assigned patients
- Patient login — patients can view their own data only
- Third-party app access — research applications can request scoped access to anonymized data
- Token-based API security — all FHIR endpoints require a valid OAuth 2.0 bearer token

11. FHIR Subscriptions — Real-Time Alerts

FHIR R4 introduces a Subscription resource that allows clients to register for notifications when specific data changes [3]. This is particularly valuable for the Smartband V001 emergency alert system.

Subscription Trigger	Notification Target	Clinical Purpose
Heart Rate > 100 BPM	Doctor's phone / EHR system	Tachycardia alert
Body Temperature > 38.5°C	Doctor's phone / EHR system	Fever alert
Acceleration > 2.0G (fall)	Emergency services API	Fall detection alert
SpO2 < 95%	Doctor's phone / EHR system	Hypoxia alert
No movement for 8+ hours	Carer notification system	Inactivity / wellness check
Battery < 10%	Patient's phone	Device maintenance alert

12. Integration with External EHR Systems

A key long-term goal of the FHIR implementation is to enable Smartband V001 data to flow directly into hospital Electronic Health Record (EHR) systems [6]. Since FHIR is the standard interface for most modern EHRs, this becomes straightforward once the FHIR API is in place.

EHR System	FHIR Support	Integration Method
Epic	FHIR R4 + SMART on FHIR	OAuth 2.0 app registration, FHIR API calls
Cerner	FHIR R4 + SMART on FHIR	Millennium FHIR API
OpenMRS	FHIR R2/R4	Open source, self-hosted integration
KIS (German)	HL7 v2 / FHIR R4	FHIR facade over existing HL7 v2 system
Meditech	FHIR R4	Expanse FHIR API

For the HSHL academic context, integration with an open-source EHR such as OpenMRS or a HAPI FHIR test server is the most practical first step, allowing full demonstration of EHR integration without requiring access to a production hospital system.

13. Conclusion and Recommendations

The integration of HL7 FHIR R4 standards into the Smartband V001 project represents a significant opportunity to transform a student wearable prototype into a clinically meaningful, interoperable health monitoring device [1]. The combination of the hardware sensors already present on the Band01_Rev1 PCB, the BLE communication infrastructure, and the Raspberry Pi gateway creates a solid foundation on which a complete FHIR implementation can be built.

The recommended next steps, in order of priority, are:

9. Activate the firmware sensor reads on Band01_Rev1 to replace simulated data with real physiological measurements.
10. Expand the FHIR API to include SpO2, step history, and alert Condition resources.
11. Deploy a HAPI FHIR server instance and migrate from the local Flask implementation.
12. Implement SMART on FHIR authentication to secure the API.
13. Integrate FHIR Subscriptions for real-time emergency alert delivery.
14. Evaluate EU MDR classification requirements for the device.
15. Conduct FHIR conformance testing using the official HL7 FHIR validator.

When fully realized, the Smartband V001 FHIR implementation will position the device to interface directly with hospital EHR systems, clinical research platforms, and consumer health applications — fulfilling the core mission of the project as defined by the Pinheiro et al. (2022) Multi-Sensor Wearable Health Device framework [1].

14. References

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